16 The probability that Saira walks to school is 0.8

If Saira walks to school, the probability that she will be late is 0.3
If Saira does **not** walk to school, the probability that she will be late is 0.05

Work out an estimate for the number of days that Saira will be late on the next 20 school days.

$P(\text{walk and late}) = 0.8 \times 0.3 = 0.24$

1 mark
Multiply along paths.

$P(\text{not walk and late}) = 0.2 \times 0.05 = 0.01$

$P(\text{late}) = 0.24 + 0.01 = 0.25$

1 mark
Add disjoint paths.

$\text{Expected days} = 20 \times 0.25 = 5$

1 mark
5 days.

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(Total for Question 16 is 3 marks)